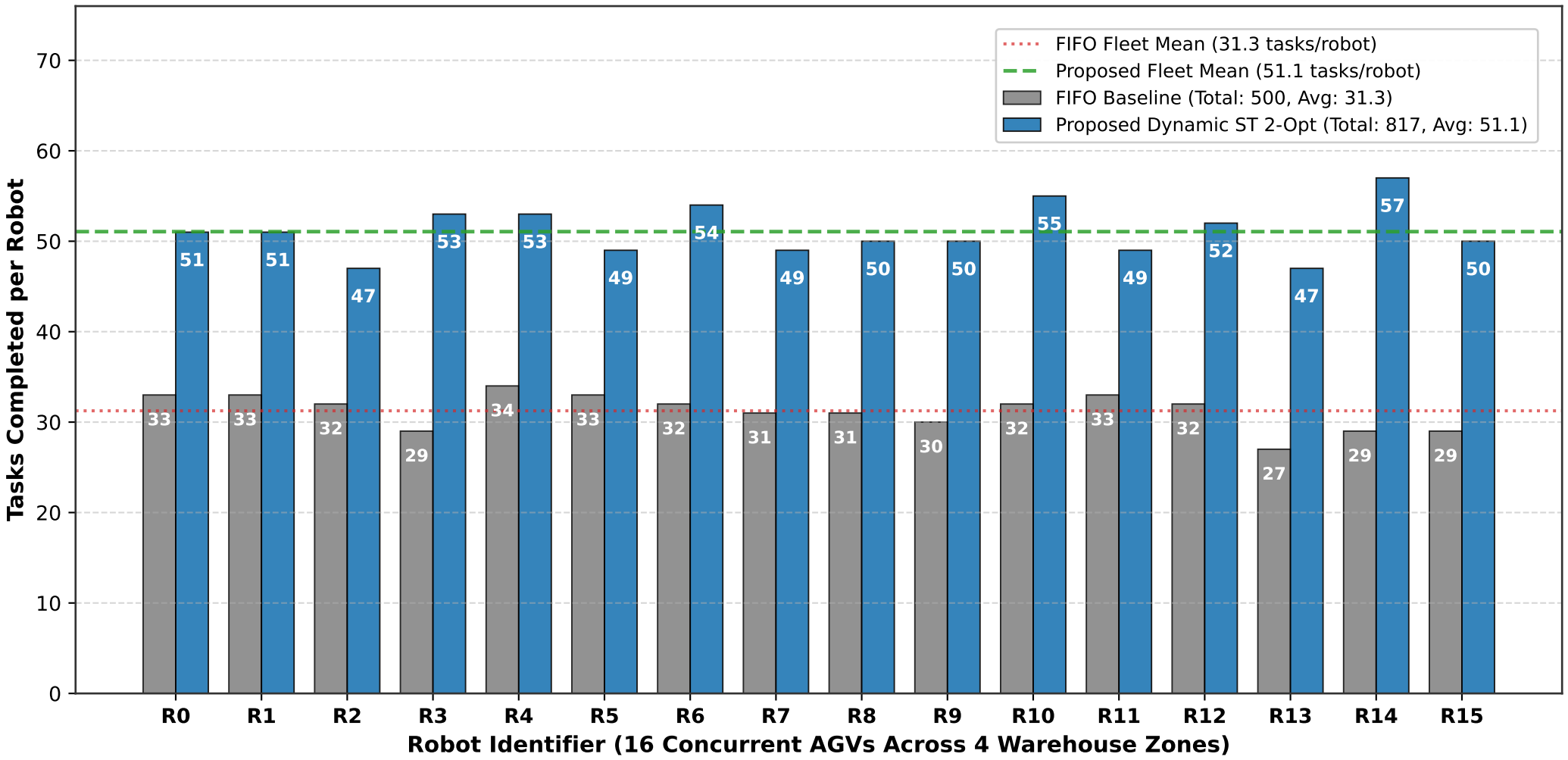
Plot 5: Fleet Workload Balance & Fairness — Per-Agent Productivity Comparison
FIFO Fleet Mean (31.3 tasks/robot)
Proposed Fleet Mean (51.1 tasks/robot)
FIFO Baseline (Total: 500, Avg: 31.3)
Proposed Dynamic ST 2-Opt (Total: 817, Avg: 51.1)
Tasks Completed per Robot
0
10
20
30
40
50
60
70
33
51
33
51
32
47
29
53
34
53
33
49
32
54
31
49
31
50
30
50
32
55
33
49
32
52
27
47
29
57
29
50
R0
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
R15
Robot Identifier (16 Concurrent AGVs Across 4 Warehouse Zones)